

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Subject: 394173 Mathematics III

Date: 7 October 2014

Year: 2014

Section: 15-16

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks and dictionary are **NOT** allowed.
4. Calculators are **NOT** allowed in the examination.
5. This is a closed book examination.
6. Electronic communication device are **NOT** allow in the examination room.
7. The examination has 9 pages (including this page), 8 questions and 1 part.
The total score is 35 points.
8. Write all your answers on this examination sheet.

No.	score
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
Total	

1. Assume that $z_1 = \frac{5i^{15}}{1-i^{87}}$ and $z_2 = \frac{4i^{45} + 8i^{2014}}{2i}$. Evaluate $\left| \frac{\overline{z_2}(z_1)^2}{z_1} \right|$.

(5 points)

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2. Find the sum of absolute value of solutions to $z^4 - 7z^3 + 15z^2 + 41z - 50 = 0$.

(4 points)

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3. Find the 3rd roots of $3\sqrt{3}i^{24} - 3i^3 - 4i^{21} + 2\sqrt{3}i^{14}$.

(5 points)

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4. Let $A = \begin{bmatrix} 1 & 0 \\ -2 & -1 \\ 2 & 3 \end{bmatrix} \bullet \begin{bmatrix} -1 & -2 & -3 \\ 1 & 2 & 1 \end{bmatrix}$ and $2B^T = \begin{bmatrix} -4 & -8 & -2 \\ 2 & -4 & 0 \\ 2 & 0 & -4 \end{bmatrix}$.

Find the skew-symmetric matrix of $A + B$.

(4 points)

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5. Given that $\begin{vmatrix} 1 & -2 & 1 \\ 2 & -2 & -1 \\ 3 & 1 & 5 \end{vmatrix} = \begin{vmatrix} x & 0 & -x \\ 3 & 0 & 2 \\ -4 & 1 & 1 \end{vmatrix}$ and $B = \begin{bmatrix} 0 & 2 & 0 & x \\ 0 & 1 & -1 & 2 \\ 0 & -x & 3 & -2 \\ 2 & -1 & -x & 4 \end{bmatrix}$.

Determine the value of $\det\left(\frac{2B^2}{B^T}\right)$.

(4 points)

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6. Evaluate B^{-1} , where $A = \begin{bmatrix} -2 & 1 & 2 \\ 3 & -1 & 0 \\ -3 & 1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$

such that $AB = C$.

(5 points)

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7. Find the value of $5p - q - 3rs$ by using Cramer's rule:

(4 points)

$$p - 2r + 3s = 1$$

$$2q - p - r + 3s = 0$$

$$3s - r - p = 0$$

$$2p + r + 3s = r.$$

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8. Solve the system of equations using Gauss–Jordan elimination:

(4 points)

$$2x_1 + 2x_2 + 4x_3 - 2x_4 = 8$$

$$3x_2 - x_3 + 4x_4 = 2$$

$$x_1 + 2x_2 - 3x_3 + 5x_4 = 0$$

$$x_1 + x_2 - 5x_3 + 6x_4 = -3$$

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